Games and Information

Midterm Examination — Spring 2023

Prof. M. Brandt

Thursday, March 9, 2023 · 80 minutes · 60 points

Name: _____ Student ID: _____

Instructions.

- There are 4 problems worth 15 points each. Answer all of them in the blue book.
- You may use one double-sided page of handwritten notes. No calculators, phones, or other electronic devices.
- Show your reasoning. A correct number with no justification receives little credit; a clearly explained approach with an arithmetic slip receives most of the credit.
- Unless stated otherwise, "equilibrium" in a game of incomplete information means Bayesian Nash equilibrium (BNE) for simultaneous-move games and perfect Bayesian equilibrium (PBE) for signaling games. You may leave answers as fractions.

Problem	1	2	3	4	Total
Points	15	15	15	15	60
Score					

Problem 1: Cournot with a privately known cost (15 points)

Two firms produce a homogeneous good and choose quantities $q_1, q_2 \geq 0$ simultaneously. Inverse demand is

$$P(q_1 + q_2) = 120 - q_1 - q_2.$$

Firm 1 has constant marginal cost $c_1 = 30$, and this is common knowledge. Firm 2's constant marginal cost c_2 is its private information: $c_2 = 18$ ("low" type, L) with probability $1/3$ and $c_2 = 36$ ("high" type, H) with probability $2/3$. Firm 1 knows only these probabilities. There are no fixed costs.

(a) Describe this situation as a Bayesian game: list the players, their type sets, their action sets, their beliefs, and what a (pure) strategy is for each firm. **[3 pts]**

(b) Derive the best response of each type of firm 2 to a quantity q_1, and firm 1's best response to a strategy (q_2^L, q_2^H) of firm 2. **[4 pts]**

(c) Find the Bayesian Nash equilibrium. Report q_1^*, q_2^{L*}, q_2^{H*} and the market price in each state. **[4 pts]**

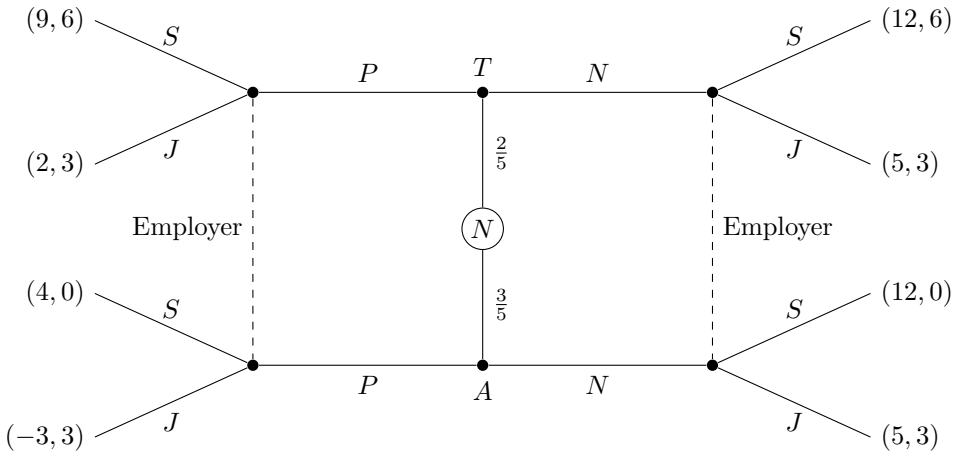
(d) Now suppose firm 1 learns c_2 before choosing its quantity, and firm 2 knows this (so each state is a standard Cournot game with complete information). Find the equilibrium quantities in each state. Compare the high-cost type's output with part (c) and explain the direction of the difference. How much (in expected profit) is this information worth to firm 1? **[4 pts]**

Problem 2: Signaling with a portfolio (15 points)

A job candidate (the sender) is either *Talented* (T) with probability 2/5 or *Average* (A) with probability 3/5. The candidate knows her type; the employer (the receiver) does not. The candidate first decides whether to submit a costly work portfolio (P) or not (N). The employer observes this choice (but not the type) and then offers either a *Senior* position (S) or a *Junior* position (J).

- The candidate's salary is 12 in a Senior position and 5 in a Junior position. Preparing a portfolio costs 3 for a Talented candidate and 8 for an Average candidate.
- The employer's payoff is 6 from placing a Talented candidate in a Senior position, 0 from placing an Average candidate in a Senior position, and 3 from any Junior hire. The portfolio has no direct value to the employer.

The game tree is shown below; payoffs are (candidate, employer).



Throughout, restrict attention to pure-strategy PBE.

(a) Let μ denote the employer's belief that the candidate is Talented after some message. For which values of μ is S a best response? **[2 pts]**

(b) Find all separating PBE. For each, state both players' strategies and the employer's beliefs, and verify that no type wants to deviate. **[5 pts]**

(c) Find all pooling PBE. Be precise about the off-path beliefs that support each one. **[4 pts]**

(d) Does the pooling equilibrium you found survive the Intuitive Criterion? Explain briefly. **[2 pts]**

(e) Suppose the Average type's portfolio cost falls from 8 to 6 (everything else unchanged). Show that no separating PBE exists. **[2 pts]**

Problem 3: A small auction (15 points)

A seller auctions one antique clock to two bidders. Bidder i's value v_i (in hundreds of dollars) is private information and is drawn independently from the uniform distribution on $[2, 6]$. Bidders

are risk-neutral; a bidder who does not win pays nothing and gets 0. Ties occur with probability zero and can be ignored.

(a) In a second-price sealed-bid auction, explain why bidding one's value is a weakly dominant strategy. Compute the seller's expected revenue. **[3 pts]**

(b) In a first-price sealed-bid auction, look for a symmetric BNE in which each bidder uses a strategy of the form $b(v) = \alpha + \beta v$. Show that $b(v) = 1 + v/2$ and verify that it is a best response for every type. **[5 pts]**

(c) Compute the seller's expected revenue in the first-price auction. **[3 pts]**

(d) Compute the expected payment of a bidder whose value is $v = 5$ under each format. Which general result do parts (a)–(d) illustrate? **[4 pts]**

Hint: if X and Y are independent $U[a, b]$, then $\mathbb{E}[\max\{X, Y\}] = a + \frac{2}{3}(b - a)$ and $\mathbb{E}[\min\{X, Y\}] = a + \frac{1}{3}(b - a)$.

Problem 4: Lemons and certification (15 points)

In a used-car market, each car has quality θ drawn uniformly from $[0, 12]$ (thousands of dollars). There is a large number of sellers, each owning one car and knowing its quality; there is a larger number of identical, risk-neutral buyers who do *not* observe quality. A seller of quality θ values keeping her car at θ; any buyer values a car of quality θ at $\theta + 3$. All cars sell at a single market price p, and competition among buyers drives p to buyers' expected value of the cars actually offered for sale.

(a) Which sellers offer their car at price p? Find the equilibrium price and the fraction of cars that are traded. **[4 pts]**

(b) With full information every car would be traded. How much expected surplus (per car in the population) is lost because of asymmetric information? **[3 pts]**

(c) Now suppose instead that buyers value a car of quality θ at $1.5\,\theta$ (sellers still value it at θ). Show that the market unravels completely. **[3 pts]**

(d) Return to the original valuations (θ for sellers, $\theta + 3$ for buyers). A garage now offers certification: for a fee of 1 paid by the seller, the car's true quality becomes publicly verifiable, and a certified car of quality θ sells for $\theta + 3$. Uncertified cars sell at a single price p_u equal to buyers' expected value of uncertified cars offered. Find the equilibrium: which sellers certify, what is p_u, and are all cars now traded? **[5 pts]**

— End of exam —